

Partial fraction Decomposition

Introduction

Suppose we have $\frac{2}{3x} + \frac{4}{5y}$ How can we combine them into a single Fraction?

$$\frac{(5y)}{(5y)} \frac{2}{3x} + \frac{4}{5y} \frac{(3x)}{(3x)} \rightarrow \frac{10y}{15xy} + \frac{12x}{15xy} \rightarrow \frac{10y + 12x}{15xy} \quad \checkmark$$

Partial fraction is the reverse problem

Partial fraction:

$$\frac{10y + 12x}{15xy} \rightarrow \frac{2}{3x} + \frac{4}{5y}$$

Example

$$\frac{7x-23}{x^2-7x+10}$$

① factor the fraction completely
(linear factors, quadratic factors)

2) separate Factors

3) operate, to delete any denominator

4) Solve For the variables
(A, B, C, ...)

linear factor
 $x+2, 3x-5$

quadratic factors
 x^2+8x+3, x^2+7

$$1) \frac{7x-23}{(x-5)(x-2)}$$

$$x^2 - 7x + 10$$

two num that + = 7
and $x = 10 > 5, 2$

$$2) \frac{7x-23}{(x-5)(x-2)} = \frac{A}{x-2} + \frac{B}{x-5}$$

$$3) 7x-23 = \left[\frac{A}{x-2} + \frac{B}{x-5} \right] (x-5)(x-2)$$

$$7x-23 = \frac{A(x-5)\cancel{(x-2)}}{\cancel{x-2}} + \frac{B\cancel{(x-5)}(x-2)}{\cancel{x-5}}$$

$$7x-23 = A(x-5) + B(x-2)$$

$$4) 7x - 23 = A(x-5) + B(x-2)$$

Solve for A & B

let $x=2 \rightarrow$ Solve A

$$7(2) - 23 = A(2-5) + B(\cancel{2-2})$$

$$14 - 23 = -3A$$

$$-9 = -3A \rightarrow \frac{-9}{-3} = A$$

$$A = 3$$

let solve $x=5 \rightarrow$ Solve B

$$7(5) - 23 = A(\cancel{5-5}) + B(5-2)$$

$$35 - 23 = B(3)$$

$$\frac{12}{3} = B \rightarrow B = 4$$

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Solution: ✓

$$\frac{7x-23}{(x-5)(x-2)} = \frac{A}{x-2} + \frac{B}{x-5}$$

$$\frac{7x-23}{(x-5)(x-2)} = \frac{3}{x-2} + \frac{4}{x-5}$$